

Postman README: Standard Postman Collection

April 2023

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Purpose

This document has been created by the Aravo Integrations Team to aid users in their use of the Standard Postman Collection. The Standard Postman Collection presents each of the supported web services Aravo provides. The Postman Collection references this document within each service. The reference will point to one of the sections and use cases within the Table of Contents.

Many web services outlined in this collection are SOAP formatted web services. These SOAP web services use XML requests and return XML responses. This represents the current day standard for the Aravo APIs. Specific web services are available in REST format and are included in this collection.

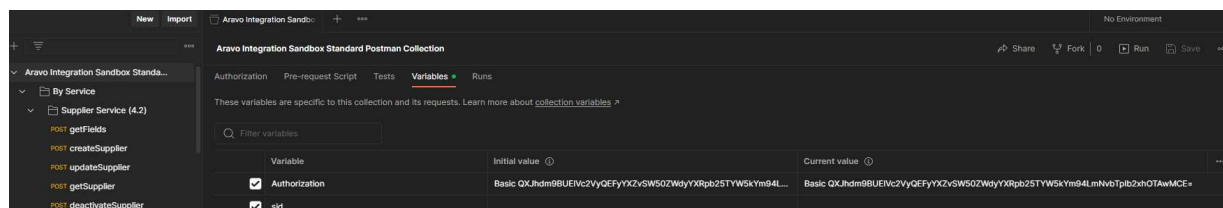
Please direct all feedback and questions regarding the Standard Postman Collection and this document to psintegrations@aravo.com. If API credentials have not been provided, or if you need them refreshed, please request this from psintegrations@aravo.com.

This document and the Postman collection are populated with notes, constraints, and/or data points to ensure the user's Postman collection/environment is set up correctly. **The approach for each section is to follow the guidance sequentially.** If testing is out of order, please refer to previous section points to ensure all information has been configured and populated correctly.

General Documentation Notes

1. JavaScript automatic variables have been included. These variables grab relevant IDs from previous responses for use in following requests. All JavaScript variables will be identifiable by the surrounding double brackets, for example "{{someVariable}}". You can view the values of these variables under the parent folder "Standard Postman Collection" > "Variables". If you wish to input a value manually, simply remove the variable from the request body such as "{{sidUseCase1}}" with the supplierId value, such as "1234567".
2. Content-Type and Endpoints are provided.
3. Credentials have been provided for this collection and are already set to be referenced in each request.

4. Please verify your credentials have been provided by viewing the folders “Standard Postman Collection” > “Variables”.



5. For supplier creation requests, unique names and identifiers are generated automatically, specific to your organization.
 - a. These are found in the “createSupplier” calls, shown as “{{sampleSupplierName}}” and “{{sampleSupplierIdentifier}}” in the request body. Please ensure you update the “client” value in the same location as #4 if you wish to dynamically generate suppliers.
 - b. These variables automatically iterate upwards, allowing for unique creation each time the request is sent.
 - c. The default format for these variables are as follows:


```

{{sampleSupplierName}}    Aravo_API_Demo_OrgName_01++
{{sampleSupplierIdentifier}} Aravo_API_Demo_OrgName_Identifier_01++
          
```
6. Default attributes/fields have been included in this collection for testing purposes. All requests are complete and can be run.
7. The collection has been organized into two sections, one which is organized by the specific web service with all relevant requests, the other is organized by potential business use cases.

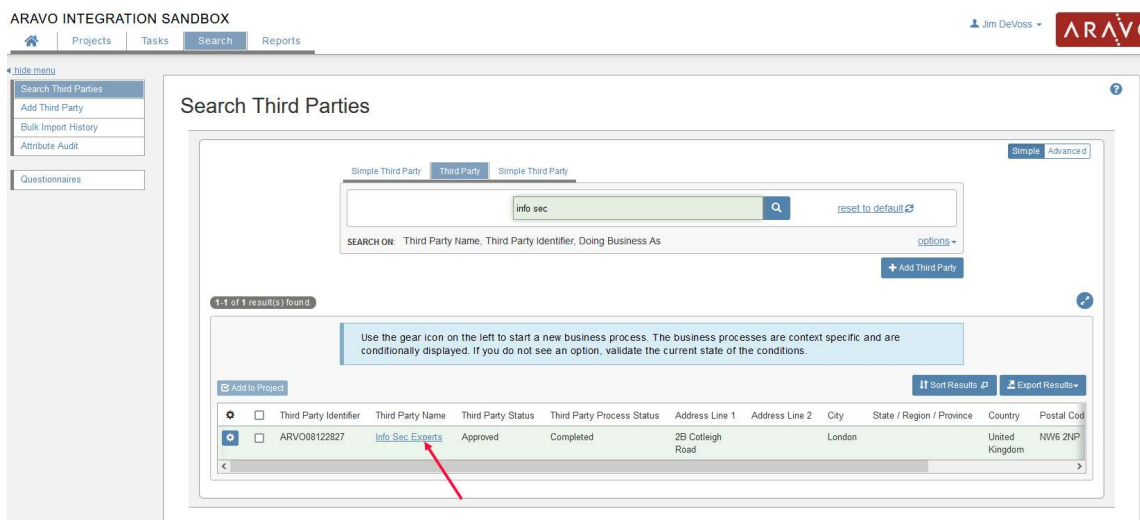
SECTION 1: Requests Organized by Service

Supplier Service (4.2) – General Notes

1. The Supplier Service facilitates the creation, modification, and retrieval of information at the “Third Party” AKA “Supplier” level. The Supplier entity is the “parent” entity of all other entity types, meaning all other entity types will exist under a supplier and can be observed by reviewing the supplier.
2. Within the Aravo user interface, the Supplier entity can be searched by name, identifier, or the “doing business as” name.



- After selecting a supplier that was modified/created, changes can be observed by reviewing the “Attribute Audit” against the supplier. First, search a specific supplier which was modified with the Postman Collection. Next, click the “Third Party Name” link.



- After entering the supplier information page, select “Attribute Audit” from within the left-nav options.

ARAVO INTEGRATION SANDBOX

Home Projects Tasks Search Reports

hide menu

- Third Party Information
 - Supplier Information
 - Supplier 360° View
 - Third Party 360° View
 - Business Relationships
 - Evaluate Dashboard
 - Engagements
 - Assessments
 - Contacts
 - Issues & Corrective Actions
 - Banks
 - Incidents
 - Files
 - SCR Hazards
 - Notes
 - Additional Sites
- Questionnaires
- Attribute Audit
- Projects
- Email
- Processes
- Surveys

History

- Attribute Audit
- Email
- Processes
- Surveys

Third Parties > Third Party Information

IDENTIFIER: ARV008122827

Info Sec Experts

Third Party Information

Third Party

Total Inherent Risk Score

Third Party Status

Third Party Type

Third Party Name

Third Party Identifier

Self Service Portal Enabled

Relationship Manager

- If the changes against the supplier were made within the last week, then the attribute audit timeframe can be left as default. All changes made will be reflected in this audit.

Third Parties > Third Party Information > Attribute Audit History

IDENTIFIER: ARV008122827

Info Sec Experts

Attribute: -- All attributes --

Context: -- All context --

Changes From: 09/28/2022 (mm/dd/yyyy) To: 04/04/2023 (mm/dd/yyyy)

☐ Include changes to pending data

Search

137 Attribute Changes

Id	Timestamp	User	User Account	User Organization	Audit Entry Type	Object	Property	Old Value	New Value	Data State	Context
1	03/09/2023 at 06:07:59 PM GMT	Mousavi, Marcus	mmousavi@aravointegration.buyer	Aravo Integration Sandbox	Update	Third Party	Public Altman Z-Score	1.4932333333333334		Committed	Import Job 15980138
2	03/09/2023 at 06:07:59 PM GMT	Mousavi, Marcus	mmousavi@aravointegration.buyer	Aravo Integration Sandbox	Add New					Committed	Import Job 15980138
3	03/09/2023 at 06:07:59 PM GMT	Mousavi, Marcus	mmousavi@aravointegration.buyer	Aravo Integration Sandbox	Add New		zzzLocation			Committed	Import Job 15980138

Supplier Service (4.2) – Provided API Calls

1. “getFields” will return all available attributes in the Data Dictionary for the Supplier entity. Direct Data Dictionary access is not included with trial access, but test attributes have been included in all following requests.
2. “createSuppliers” will create a new supplier. This supplier can be found in the Aravo UI by following the steps outlined in “Supplier Service (4.2) – General Notes” #2 - #3.
3. “updateSuppliers” will update an existing supplier by referencing the previously returned “supplier.id”.
 - a. The JavaScript variables will take care of transferring this value between requests within the collection.
4. “getSuppliers” will fetch the data for a specified supplier.
5. “deactivateSuppliers” will mark an existing supplier as inactive. Inactive supplier’s cannot be interacted with via the API once they are deactivated. Manual reactivation within the UI is needed to reestablish API eligibility access to the supplier record.

Supplier Contact Service (4.2) – General Notes

1. The Supplier Contact Service facilitates the creation, modification, and retrieval of information at the Contact entity level.
2. Contacts are added to a specified supplier upon creation by referencing the {{sid}} aka “supplier.id”.
3. The JavaScript variables “{{someVariable}}” will take care of transferring values from previous API calls to these requests within the collection.

Supplier Contact Service (4.2) – Provided API Calls

1. “getFields” will return all available fields in the Data Dictionary for the Contact entity.
2. “createSupplierContacts” will create a new contact.
3. “updateSupplierContacts” will update an existing contact.
4. “getSupplierContacts” will fetch the data for a specified contact.
5. “deactivateSupplierContacts” will mark an existing contact as inactive.

Business Relationship Service (4.2) – General Notes

1. The Business Relationship Service facilitates the creation, modification, and retrieval of information for entity types of “Business Relationship”.
2. Business Relationship entities can be customer defined to encompass a wide variety of different use cases.

- Some examples of a customer defined Business Relationships would be “Site”, “Tax Documents”, and “Contract”.
- The Business Relationship service references the entity key, as specified in the Data Dictionary, to create, update, and retrieve data. Please see section “brType/Business Relationship Type” within Section 3 below.

The screenshot shows the Aravo Data Dictionary interface. On the left is a sidebar menu with options: Overview, Data Dictionary (highlighted), Forms, Workflows, Searches, User Interface, Evaluate, Projects, Supplier Portal, Advanced, Configuration Management, Scripts, and Aravo AI. The main content area is titled 'Data Dictionary' and has tabs for Extensible Objects, Text Fields, Conditions, Validations, Selection Lists, Selection Hierarchies, and Filters. The 'Extensible Objects' tab is active. It prompts the user to 'Please select an entity to edit:' with a dropdown menu showing 'Engagement' and a link to 'view entity graph'. Below this, the 'Entity Details' section shows: Name: Engagement, Description: Engagement, Type: Business Relationship, Key: TPMEngagement (highlighted with a red bar), and Cloning: edit. There are links for 'edit' and 'hide' for the details. The 'Associations' section has a 'show' link. The 'Attributes' section has a 'hide' link. At the bottom, it says 'Click column header to sort, click row to edit'.

- For this example, the Business Relationship type entity “Engagement” was used.
- The JavaScript variables “{{someVariable}}” will take care of transferring values from previous API calls to these requests within the collection.

Business Relationship Service (4.2) – Provided API Calls

- “getFields” will return all available fields in the Data Dictionary for the specified entity.
- “createBusinessRelationships” will create a new entity of the specified type.
- “updateBusinessRelationships” will modify an existing entity.
- “getBusinessRelationships” will retrieve an existing entity record by referencing both the supplier and business relationship IDs.
- “getBusinessRelationships2” will retrieve an existing entity record by referencing only the business relationship ID.
- “deactivateBusinessRelationships” will mark an existing entity as inactive.

Association Service (4.2) – General Notes

- The Association Service is used to create, update, and retrieve a referenceable relationship between two entities.
- Associations can be customer defined in a wide variety of ways, including 1:1, 1:N, and N:N relationship structure.

3. The field “associationKey” refers to the Association Key as defined in the Data Dictionary.

The screenshot shows the Aravo Data Dictionary interface. On the left is a sidebar with navigation links: Overview, Data Dictionary (selected), Forms, Workflows, Searches, User Interface, Evaluate, Projects, Supplier Portal, Advanced, Configuration Management, Scripts, and Aravo AI. The main area displays a table of associations with columns for Name, Description, and Association Definition. The table lists several associations, including 'Financial Health Assessment - Supplier', 'Supplier - CQuayEngagement1', 'Supplier - CQuayEngagement2', 'Supplier - CQuayEngagement3', 'Supplier - CQuayTest', 'Supplier - CQuayTest2', 'Supplier - Corrective Action', 'Supplier - hTorTestEnt', 'Supplier_Contact', and 'Supplier_Engagement'. An 'Association Details' modal is open for the 'Supplier_Engagement' association. The modal has tabs for Basic Info, Form Fields, Rollup, and Usage. The 'Basic Info' tab is active, showing the Name 'Supplier_Engagement', a Description field, and the Association Definition. The Association Definition shows a diagram with 'Supplier' and 'Engagement' entities. 'Supplier' is connected to 'Engagement' via an 'A-B Relationship' (labeled 'A' to 'B') and a 'B-A Relationship' (labeled 'B' to 'A'). The 'Supplier' entity is marked as the primary engagement. The Association Key is 'Supplier_Engagement'. The Association ID is 774121. Buttons for Save, Cancel, and Delete Association are at the bottom.

4. Associations have the functionality of setting a “Primary” flag for distinguishing between multiple entities of the same type if the “Primary” setting has been enabled for the association in the Data Dictionary.

This is a close-up of the 'Association Details' modal for the 'Supplier_Engagement' association. The 'Basic Info' tab is selected. The Name is 'Supplier_Engagement'. The Description field is empty. The Association Definition section shows a diagram with 'Supplier' and 'Engagement' entities. 'Supplier' is connected to 'Engagement' via an 'A-B Relationship' (labeled 'A' to 'B') and a 'B-A Relationship' (labeled 'B' to 'A'). The 'Supplier' entity is marked as the primary engagement. The Association Key is 'Supplier_Engagement'. The Association ID is 774121. Buttons for Save, Cancel, and Delete Association are at the bottom.

- a. For example, a single supplier has multiple contacts, but one contact is associated with the “Primary” flag, which indicates that this is the primary contact for the supplier.

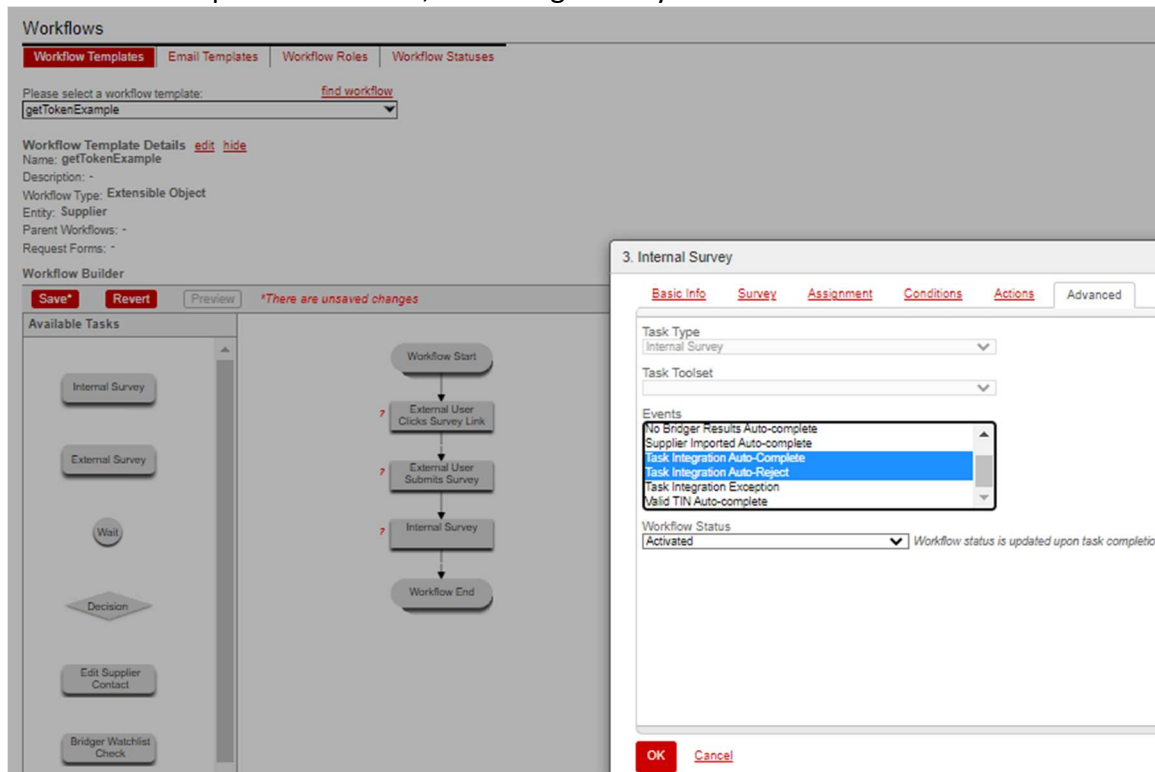
Association Service (4.2) – Provided API Calls

1. “getAssociationTypes” will return the available “Association Keys”, which are defined within the Data Dictionary.
2. “addAssociations” will allow for the creation of an association between two entities.
 - a. The left entity is the entity on the left side of the association key.
 - b. The right entity is the entity on the right of the association key.
 - c. The supplier is referenced at the bottom of the request to give context to where the association is taking place.
3. “getAssociatedEntities” will allow for the lookup of existing associations to a specific entity.
 - a. The “associationKeyFilter” allows for the reference of a specific association key.
 - b. The “subtypeKey” is the entity key related to the ID you are searching against.
4. “getAssociations” will allow for the lookup for an existing association by associationId.
 - a. The associationIds are returned after association creation or can be found by calling getAssociatedEntities.
5. “updateAssociations” will allow for the modification of an existing association.
 - a. In the provided example, a contact is being changed from non-primary status to primary status.
6. “deactivateAssociations” will allow for the deletion or marking of an inactive status for an existing association by referencing the associationId.
7. “deactivateAssociations2” will allow for the deletion or marking of an inactive status for an existing association by referencing the association key, left entity ID, and right entity ID.

Business Process Service (4.4) – General Notes

1. The Business Process Service is utilized for starting a Business Process against a supplier or other entity and retrieving information about a specific Business Process.
2. Business Processes are executions of a specific workflow.
 - a. Workflows are best understood as an outlined chain of logic for business use cases. The logic “chain” is composed of “Tasks”.
 - b. Business processes are instances of workflows that are run against a specific entity, such as a supplier.

3. The Business Process Service is also capable of completing or rejecting tasks which have been configured for this use.
4. To configure a task for completion or rejection, enable the following in the task configuration.
 - a. Please enable this configuration for only a single task at a time.
 - b. If two tasks, which occur in order, have these options enabled, a single API call will complete them both, which is generally not desired.



5. The JavaScript variables “{{someVariable}}” will take care of transferring values from previous API calls to these requests within the collection. Note that for this case, the {{bpid}} variable references either the supplier OR entity business process that was started, depending on which was last started.

Business Process Service (4.4) – Provided API Calls

1. “startSupplierBusinessProcesses” will allow for a Business Process to be started against an existing supplier by referencing the workflowTemplateName and supplierId.
2. “startEntityBusinessProcesses” call allows for a Business Process to be started against an existing entity by referencing the workflowTemplateName, entityId, and “Key” of the entity. These have been populated in the collection.
3. The “getBusinessProcessDetails” call returns the current status of a business process.

4. The “notifyIntegrationCompleted” call is used with the “Wait For External System” workflow task. This call marks the task as completed.
5. The “notifyIntegrationRejected” call is used with the “Wait For External System” workflow task. This call marks the task as rejected. The current example is not configured for rejection.

User Service (4.2) – General Notes

1. The User Service is used to create, update, and retrieve users against the specified Aravo Buyer.
2. The User Service is capable of modifying roles/permissions of specified users.
3. Additional information about assigning and removing Custom Roles, Task Assignments, Action Assignments, and more is available upon request.

User Service (4.2) – Provided API Calls

1. “getFields” will return a list of core attribute names that are accepted in create and update operations.
2. “createUsers” will allow the creation of one or more users and set values of user attributes/permissions.
3. “updateUsers” will allow the update of attributes or permissions of one or more users.
4. “getUsers1” will return a user’s information by referencing the userId.
5. “getUsers2” will return a user’s information by referencing the personId.
6. “deactivateUsers” allows the deactivation of an existing user.

Business Process REST Service – General Notes

1. The Business Process REST Service is capable of completing or rejecting tasks. It is not capable of starting a new business process or retrieving business process details like the SOAP version.

Business Process REST Service – Provided API Calls

1. The “Task Integration Auto-Complete” call is generally used with the “Wait for External System” workflow task. This call marks the task as completed.
2. The “Task Integration Auto-Complete” call is generally used with the “Wait for External System” workflow task. This call marks the task as rejected.

File Attributes REST Service – General Notes

1. The File Attributes REST Service allows for the creation, retrieval, and management of a file within Aravo on a specific record.
2. This service is required to be used in almost every single instance of file reference via API within Aravo.
3. Additional information is available upon request.

File Attributes REST Service – Provided API Calls

1. “GET” returns a filename, description, contentType, base64Bytes, revisionNumber, modifiedDate, and size of a file stored on a record within Aravo by being provided with an entityType, entityId, and attributeKey.
2. “POST” will upload a file onto a record within Aravo by providing an entityType, entityId, attributeKey, filename, description, contentType, and base64Bytes encoded file.
3. “DELETE” will delete a file on a record within Aravo by being provided with an entityType, entityId, and attributeKey.

OAuth2 Authorization Service

1. “getToken” will allow for the generation of a token with a lifespan for use in the Authorization header of other API calls.
 - a. You can exchange your Base64 encoded username and password for the token returned by this call for any other service. Note that the format would change from “Basic ASJNEJCNENCE++” to “Bearer xxxx”.

Reports Service – General Notes

1. The Reports Service is responsible for the retrieval of previously created reports.
2. Reports are highly customizable and can be configured to display almost any type of information desired.
3. Reports can be found within the Buyer under the “Reports” tab.

Reports Service – Provided API Calls

1. “getReports” will return a summary of all reports which the API user has access to.

2. “getReportsById” will return a report and all data included in that report by referencing the reportId in the URL.

SECTION 2: Requests Organized by Use Case

Use Case 1 - Creation

1. This use case is designed to demonstrate the creation of a supplier, relevant engagement, relevant contact, and association between all three. It then shows how to start a business process against the newly created supplier.
2. Start with the creation of a supplier, this is the parent of each created entity that follows. For example, you create a contact or engagement entity, both reside under the supplier entity. This means that the supplier entity must exist first.
3. The use cases are designed to be run from top to bottom so that the JavaScript variables can be populated, and the order of operations are correct. You can disregard this if you are manually populating IDs instead of relying on the variables.

Use Case 2 – Retrieval

1. This use case is designed to demonstrate how to retrieve the data created in the first use case.
2. These requests can be run in any order, the IDs were populated in the JavaScript variables when Use Case 1 was run.
3. You can replace the JavaScript variables at any time with valid IDs.

Use Case 3 – Update

1. This use case is designed to demonstrate how to update the suppliers, engagements, contacts, and associations which were created in the first use case.
2. These requests can be run in any order, the values were populated in the first use case.
3. You can replace the JavaScript variables at any time with valid IDs.

Batching Example

1. This use case is designed to demonstrate how to batch multiple requests together to avoid excessive API calls.
2. Each request in this example starts with a format similar to “<createRequest>” and ends with “</createRequest>”.

OAuth2 + Reports Example

1. This use case is designed to demonstrate how to use OAuth2 in the Authorization header and how to use the Reports Service.
2. The response from the “getToken” call is meant to be used in the Authorization of the Reports calls as follows: “Bearer AshjasAHSKnMDJLLSJE” or “Bearer {token value}”. This has been automatically transferred for your convenience.
3. The getReports call will list all available reports to the user. The getReportsById call will fetch a single report with data.

Override Active Business Process Example

1. The field “<overrideActiveBusinessProcess>true</overrideActiveBusinessProcess>” allows you to modify a supplier or entity which is the process subject of an active business process.
2. Note that recommended use, for this field, is when the workflow is expecting entity modification, such as a “Wait for External System” Task.
3. Updating an entity during an active business process may have unintended consequences, depending on numerous factors. These factors may include: the attributes being updated (may be used in conditional logic), if a task is actively performing some logic (such as a script currently running), or if a workflow is in the process of starting child workflows (the entity profile may be locked during this process).

Clear Existing Value Example

1. When the “<values> </values>” field is not included in the API call, you are telling the system to clear the existing value.

SECTION 3: Documentation of Constraints + Examples

Maximum Number of Requests by Service – Supplier Service (4.2)

Operation	Number of Batched Requests
createSuppliers	100
updateSuppliers	100
getSuppliers	100
deactivateSuppliers	100

Maximum Number of Requests by Service – Supplier Contact Service (4.2)

Operation	Number of Batched Requests
createSupplierContacts	100
updateSupplierContacts	100
getSupplierContacts	50
getSupplierContacts2	50
deactivateSupplierContacts	100

Maximum Number of Requests by Service – Business Relationship Service (4.2)

Operation	Number of Batched Requests
createBusinessRelationships	100
updateBusinessRelationships	100
getBusinessRelationships	50
getBusinessRelationships2	50
deactivateBusinessRelationships	100

Maximum Number of Requests by Service – Association Service (4.2)

Operation	Number of Batched Requests
addAssociations	100
getAssociations	50
getAssociatedEntities	50
updateAssociations	100
deactivateAssociations	100

Maximum Number of Requests by Service – User Service (4.2)

Operation	Number of Batched Requests
createUsers	100
updateUsers	100
getUsers	50
getUsers2	50
deactivateUsers	100

Maximum Number of Requests by Service – Business Process Service (4.4)

Operation	Number of Batched Requests
startSupplierBusinessProcess	100
startEntityBusinessProcess	100
getBusinessProcessDetails	100
notifyIntegrationCompleted	100
notifyIntegrationRejected	100

Aravo Data Types in API Calls - Boolean

Types accepted: <type>bool</type>

Values: “true” or “false”

Supported Calls: All relevant services

Unsupported Calls: None

Example in API Call:

```
<fields>
  <name>ex.isNewEntity</name>
  <type>bool</type>
  <values>true</values>
</fields>
```

Aravo Data Types in API Calls – Contact Select

Types accepted: <type>code</type> or <type>username</type> or <type>id</type>

Values: Reference the pid (personId) or username of a previously created contact.

Supported Calls:

1. Supplier Service
 - a. updateSuppliers
2. Business Relationship Service
 - a. createBusinessRelationships
 - b. updateBusinessRelationships

Unsupported Calls:

1. Supplier Service
 - a. createSuppliers - contact does not exist for a supplier before creation.
 - b. getSuppliers - This type is not supported in export.
2. Business Relationship Service
 - a. getBusinessRelationships - This type is not supported in export.

Examples in API Call:

```
<fields>
  <name>ex.contactSelectExample</name>
  <type>username</type>
  <values>exampleUsername@aravo.com</values>
</fields>

<fields>
  <name>ex.contactSelectExample</name>
  <type>code</type>
  <values>35362627</values>
</fields>

<fields>
  <name>ex.contactSelectExample</name>
  <type>id</type>
  <values>35362627</values>
</fields>
```

Aravo Data Types in API Calls – Country

Types accepted: <type>iso</type> or <type>code</type> or <type>label</type>

Values: The country type attribute uses the “Country (core list)” selection list. In this case “iso” takes an input of the “ISO Code” value of this list, “code” takes an input of the “Code” value of this list, and “label” takes an input of the “Description” value of this list (shown below).

Supported Calls: All relevant services

Unsupported Calls: None

Information Given for API Call:

Displayed Locale	en_US_HIERARCHY/NAME/ISO CODE
Code	United States of America
Description	United States of America
Sort Order	-1000
ISO Code	US

Examples in API Call:

```

<fields>
  <name>ex.CountryType</name>
  <type>iso</type>
  <values>US</values>
</fields>

<fields>
  <name>ex.CountryType</name>
  <type>code</type>
  <values>United States of America</values>
</fields>

<fields>
  <name>ex.CountryType</name>
  <type>label</type>
  <values>United States of America</values>
</fields>

```

Aravo Data Types in API Calls – Country Multi Select

Types accepted: <type>code</type> or <type>iso</type>

Values: The Country Multi Select type attribute uses the “Country (core list)” selection list. In this case “iso” takes an input of the “ISO Code” value of this list and “code” takes an input of the “Code” value of this list (shown below). The type “label” is not referenceable for this data type.

Supported Calls: All relevant services

Unsupported Calls: None

Information Given for API Call:

Displayed Label	en_US_HERBERT HORNBAUM CONT TO
Code	United States of America
Description	United States of America
Sort Order	-1000
ISO Code	US

Examples in API Call:

```
<fields>
  <name>ex.countryMultiSelectExample</name>
  <type>iso</type>
  <values>US</values>
  <values>CA</values>
</fields>

<fields>
  <name>ex.countryMultiSelectExample</name>
  <type>code</type>
  <values>Afghanistan</values>
  <values>Canada</values>
</fields>
```

Aravo Data Types in API Calls – Currency

Types accepted: <type>code</type> or <type>label</type>

Values: The Currency type attribute uses the “Currency List (core list)” selection list. In this case “label” takes an input of the “Description” value of this list and “code” takes an input of the “Code” value of this list (shown below). The type “iso” was not shown to be referenceable during this analysis.

Supported Calls: All relevant services

Unsupported Calls: None

Information Given for API Call:

Code	EUR
Description	Euro
Sort Order	0
ISO Currency	EUR Euro

Examples in API Call: Note that a separate currency was used for the “label” example.

```
<fields>
  <name>ex.currencyTest</name>
  <type>code</type>
  <values>EUR</values>
</fields>

<fields>
  <name>ex.currencyTest</name>
  <type>label</type>
  <values>Canadian Dollar</values>
</fields>
```

Aravo Data Types in API Calls – Currency Single Select

Types accepted: <type>code</type> or <type>iso</type>

Values: The Currency Single Select type attribute uses the “Currency List (core list)” selection list. In this case “iso” takes an input of the “Code” value of this list and “code” takes an input of the “Code” value of this list (shown below). The type “label” was not shown to be referenceable during this analysis.

Supported Calls: All relevant services

Unsupported Calls: None

Information Given for API Call:

Code	EUR
Description	Euro
Sort Order	0
ISO Currency	EUR Euro

Examples in API Call:

```
... <fields>
...   <name>ex.currencySingleSelectExample</name>
...   <type>code</type>
...   <values>EUR</values>
... </fields>

... <fields>
...   <name>ex.currencySingleSelectExample</name>
...   <type>iso</type>
...   <values>EUR</values>
... </fields>
```

Aravo Data Types in API Calls – Currency Multi Select

Types accepted: <type>code</type> or <type>iso</type>

Values: The Currency Multi Select type attribute uses the “Currency List (core list)” selection list. In this case “iso” takes an input of the “Code” value of this list and “code” takes an input of the “Code” value of this list (shown below). The type “label” was not shown to be referenceable during this analysis.

Supported Calls:

1. Supplier Service
 - a) updateSuppliers
 - b) createSuppliers
2. Business Relationship Service
 - a) createBusinessRelationships
 - b) updateBusinessRelationships

Unsupported Calls:

1. Supplier Service
 - a) getSuppliers - This type is not supported in export.
2. Business Relationship Service
 - a) getBusinessRelationships - This type is not supported in export.

Information Given for API Call:

Code	EUR
Description	Euro
Sort Order	0
ISO Currency	EUR Euro

Examples in API Call:

```
<fields>
  <name>ex.currencyMultiSelectE</name>
  <type>iso</type>
  <values>EUR</values>
  <values>USD</values>
</fields>

<fields>
  <name>ex.currencyMultiSelectE</name>
  <type>code</type>
  <values>EUR</values>
  <values>USD</values>
</fields>
```

Aravo Data Types in API Calls – Date

Types accepted: <type>date</type>

Values: Dates in the format of yyyy-mm-dd HH:mm:ss.SSS. Dates can be submitted with partial values working from left to right, but they will be auto populated for missing portions by default. Example: If I submit the value “2021” then the system will auto populate this date as “2021-01-01T00:00:00-06:00”. You must start with the leftmost value (year) and work right for submitting partial dates. This means if you have the year and a day, but not a month, then you cannot submit the day without error.

Supported Calls: All relevant services

Unsupported Calls: None

Example in API Call:

```
.....<fields>
.....  <name>ex.DateSample</name>
.....  <type>date</type>
.....  <values>2021-11-06T18:45:58.000-08:00</values>
.....</fields>
```

Aravo Data Types in API Calls – Dynamic URL + E-Signature

Types accepted: None

Values: None

Supported Calls: None

Unsupported Calls: All

Aravo Data Types in API Calls – File Entity + File Encrypted + File List + File List Encrypted

Not applicable for SOAP Services. Please see

https://prod.aravo.com/aems/apidocs/restservices/v1_0/fileattributes/

OR File Attribute REST Service in Readme

Aravo Data Types in API Calls – Hierarchy Single Select

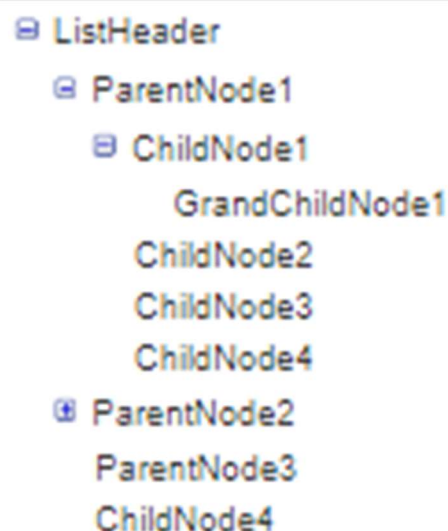
Types accepted: <type>code</type> or <type>label</type>

Values: View the image of the Selection Hierarchy below and then view how this was submitted to the API. You can see that the delimiter between nodes is “/”. For the “label” type, for each referenced node I entered a description with a number. Example: ListHeader description was 1, ParentNode1 description was 2, ChildNode1 description was 3, and GrandChildNode1 description was 4. See how this was used in practice in the “label” example below.

Supported Calls: All relevant services

Unsupported Calls: None

Information Given for API Call:



Examples in API Call:

```

<fields>
  <name>ex.hierarchySing</name>
  <type>code</type>
  <values>ListHeader/ParentNode1/ChildNode1/GrandChildNode1</values>
</fields>

<fields>
  <name>ex.hierarchySing</name>
  <type>label</type>
  <values>1/2/3/4</values>
</fields>

```

Aravo Data Types in API Calls – Hierarchy Multi Select

Types accepted: <type>code</type> or <type>label</type>

Values: View the image of the Selection Hierarchy below and then view how this was submitted to the API. You can see that the delimiter between nodes is “/”. For the “label” type, for each referenced node I entered a description with a number. Example: ListHeader description was 1, ParentNode1 description was 2, ChildNode1 description was 3, and GrandChildNode1 description was 4. See how this was used in practice in the “label” example below.

Supported Calls: All relevant services

Unsupported Calls: None

Information Given for API Call:

```
ListHeader
├── ParentNode1
│   ├── ChildNode1
│   │   ├── GrandChildNode1
│   │   ├── ChildNode2
│   │   ├── ChildNode3
│   │   └── ChildNode4
│   └── ParentNode2
│       ├── ChildNode1
│       ├── ChildNode2
│       ├── ChildNode3
│       ├── ChildNode4
│       ├── ParentNode3
│       └── ChildNode4
```

Examples in API Call:

```
<fields>
  <name>ex.hierarchyMulti</name>
  <type>code</type>
  <values>ListHeader/ParentNode1/ChildNode1/GrandChildNode1</values>
  <values>ListHeader/ParentNode2/ChildNode2</values>
</fields>

<fields>
  <name>ex.hierarchyMulti</name>
  <type>label</type>
  <values>1/2/3/4</values>
  <values>1/5/6</values>
</fields>
```

Aravo Data Types in API Calls – Multi Select

Types accepted: <type>code</type> or <type>label</type>

Values: In this case “label” takes an input of the “Description” value of this selection list and “code” takes an input of the “Display Text” value of this selection list (shown below).

Supported Calls: All relevant services

Unsupported Calls: None

Information Given for API Call:

Display Text	Description
test1	1
test2	2

Examples in API Call:

```
...<fields>
...  <name>ex.multiSelectTest</name>
...  <type>code</type>
...  <values>test1</values>
...  <values>test2</values>
...</fields>

...<fields>
...  <name>ex.multiSelectTest</name>
...  <type>label</type>
...  <values>1</values>
...  <values>2</values>
...</fields>
```

Aravo Data Types in API Calls – Single Select

Types accepted: <type>code</type> or <type>label</type>

Values: In this case “label” takes an input of the “Description” value of this selection list and “code” takes an input of the “Display Text” value of this selection list (shown below).

Supported Calls: All relevant services

Unsupported Calls: None

Information Given for API Call:

Display Text	Description
test1	1
test2	2

Examples in API Call:

```
.....<fields>
.....  <name>ex.singleSelectEx</name>
.....  <type>code</type>
.....  <values>test1</values>
.....</fields>

.....<fields>
.....  <name>ex.singleSelectEx</name>
.....  <type>label</type>
.....  <values>1</values>
.....</fields>
```

Aravo Data Types in API Calls – Number

Types accepted: <type>number</type>

Values: Either a whole number or a number with decimals included (shown below). Character limit current is: 15 (only 2 decimals places allowed), this is subject to change.

Supported Calls: All relevant services

Unsupported Calls: None

Examples in API Call:

```
.....<fields>
.....  <name>ex.test123</name>
.....  <type>number</type>
.....  <values>12345</values>
.....</fields>

.....<fields>
.....  <name>ex.test123</name>
.....  <type>number</type>
.....  <values>12345.12</values>
.....</fields>
```

Aravo Data Types in API Calls – Text

Types accepted: <type>string</type>

Values: Text up to 256 characters (current limit and subject to change)

Supported Calls: All relevant services

Unsupported Calls: None

Example in API Call:

```
.....<fields>
.....  <name>ex.StringSample</name>
.....    <type>string</type>
.....    <values>Hello World Check One</values>
.....</fields>
```

Aravo Data Types in API Calls – Text Area

Types accepted: <type>string</type>

Values: Text up to 3999 characters (current limit and subject to change. Recent discussions have mentioned this will be increased in the near future)

Supported Calls: All relevant services

Unsupported Calls: None

Example in API Call:

```
.....<fields>
.....  <name>ex.textAreaExample</name>
.....    <type>string</type>
.....    <values>Hello World Check One but Pretend its 3999 Characters</values>
.....</fields>
```

Aravo Data Types in API Calls – URL

Types accepted: <type>string</type>

Values: Unknown character limit, but verified up to 4000 characters.

Supported Calls: All relevant services

Unsupported Calls: None

Example in API Call:

```
.....<fields>
.....<name>ex.ws_test_supplier_url</name>
.....<type>string</type>
.....<values>https://demo.aravo.com</values>
.....</fields>
```

Aravo Data Types in API Calls – Workflow Role Single Select

Types accepted: <type>username</type> or <type>id</type>

Values: Username is the internal username value. id is the internal id of the user. The user must already have the role specified in the attribute before attempting to assign this user's value to the attribute.

```
<return>
<message>Failed to process ex.workflowSingleSelect: Error while updating extended attribute workflowSingleSelect: User with username: admintraining2332 is not in required role:
hTorWorkflowTraining</message>
```

Supported Calls:

1. Supplier Service
 - a) updateSuppliers
 - b) createSuppliers
2. Business Relationship Service
 - a) CreateBusinessRelationships
 - b) updateBusinessRelationships

Unsupported Calls:

1. Supplier Service
 - a) getSuppliers - This type is not supported in export.
2. Business Relationship Service
 - a) getBusinessRelationships - this type is not supported in export.

Example in API Call:

```
...<fields>
...  <name>ex.workflowSingleSelect</name>
...  <type>username</type>
...  <values>admintraining2332</values>
...</fields>

...<fields>
...  <name>ex.workflowSingleSelect</name>
...  <type>id</type>
...  <values>33641889</values>
...</fields>
```

Aravo Data Types in API Calls – Workflow Role Multi Select

Types accepted: <type>username</type> or <type>id</type>

Values: username is the internal username value. id is the internal id of the user. The user must already have the role specified in the attribute before attempting to assign this user's value to the attribute.

```
<return>
<message>Failed to process ex.workflowSingleSelect: Error while updating extended attribute workflowSingleSelect: User with username: admintraining2332 is not in required role:
hTorWorkflowTraining</message>
```

Supported Calls:

3. Supplier Service
 - a) updateSuppliers
 - b) createSuppliers
4. Business Relationship Service
 - a) CreateBusinessRelationships
 - b) updateBusinessRelationships

Unsupported Calls:

3. Supplier Service
 - a) getSuppliers - This type is not supported in export.
4. Business Relationship Service
 - a) getBusinessRelationships - this type is not supported in export.

Example in API Call:

```

<<<<<fields>
<<<<<<name>ex.workflowRoleMutli</name>
<<<<<<type>username</type>
<<<<<<values>admintraining2332</values>
<<<<<<values>admintraining2333</values>
<<<<</fields>
<<<<</fields>
<<<<<fields>
<<<<<<name>ex.workflowRoleMutli</name>
<<<<<<type>id</type>
<<<<<<values>33641889</values>
<<<<<<values>33641890</values>
<<<<</fields>

```

Aravo Other Types in API Calls – uidType/userId

Types accepted: <uidType>username</uidType> or <uidType>externalId</uidType> or <uidType>internalId</uidType>

Values: username is the internal username value. internalId is the internalId of the user. externalId is the external username of the user.

Examples in getUsers API Call:

```

<<<<<<user:getUsers>
<<<<<<<request>
<<<<<<<<uid>admintraining2332@demo.com</uid>
<<<<<<<<uidType>externalId</uidType>
<<<<<<</request>
<<<<<</user:getUsers>
<<<<<<user:getUsers>
<<<<<<<request>
<<<<<<<<uid>admintraining2332</uid>
<<<<<<<<uidType>username</uidType>
<<<<<<</request>
<<<<<</user:getUsers>
<<<<<<user:getUsers>
<<<<<<<request>
<<<<<<<<uid>33641889</uid>
<<<<<<<<uidType>internalId</uidType>
<<<<<<</request>
<<<<<</user:getUsers>

```

Aravo Other Types in API Calls – sidType/supplierId

Types accepted: <sidType>internalId</sidType> or <sidType>externalId</sidType>

Values: internalId is the internal supplier ID. externalId is the supplier identifier.

Examples in updateSupplier API Call:

```
...> |to ignore active business process-->
...> <sid>JDHFJSIFJHDSIFJ3845678912346</sid>
...> <sidType>externalId</sidType>
...> </updateRequest>
</sup:updateSuppliers>

...> |to ignore active business process-->
...> <sid>30042285</sid>
...> <sidType>internalId</sidType>
...> </updateRequest>
</sup:updateSuppliers>
```

Aravo Other Types in API Calls – brType/Business Relationship Type

Types accepted: Any specified by the customer data dictionary.

Values: See the below screenshot to see where this value is found, reference the “Key” of the entity. Must be of “Type” Business Relationship for this call to be applicable.

Data Dictionary

Extensible Objects	Text Fields	Conditions	Validations	Selection List
Please select an entity to edit: view entity graph				
hTorTestEnt				

Entity Details [edit](#) [hide](#)
Name: hTorTestEnt
Description: hTorTestEnt
Type: Business Relationship
Key: hTorTestEnt
Cloning: [edit](#)

Example in createBusinessRelationships API Call:

```

<soapenv:Body>
  <bus:createBusinessRelationships>
    <!--1 or more repetitions!-->
    <createRequest>
      <brType>hTorTestEnt</brType>
      <fields>
        <name>name</name>
        <type>string</type>

```

Aravo Other Types in API Calls – workflowTemplateName

Types accepted: Any specified by the customer workflows.

Values: See the below screenshot to see where this value is found, reference the “Name” of the workflow.

Information Given in Example:

Please select a workflow template: [find workflow](#)

taskCompleteTest

Workflow Template Details [edit](#) [hide](#)

Name: [taskCompleteTest](#)

Description: -

Workflow Type: Extensible Object

Entity: Supplier

Parent Workflows: -

Request Forms: [completeTaskEventTest](#)

Example in startSupplierBusinessProcess API Call:

```

<soapenv:Body>
  <bus:startSupplierBusinessProcesses>
    <request>
      <workflowTemplateName>taskCompleteTest</workflowTemplateName>
    <subject>

```

Aravo Other Types in API Calls – pidType/personId Type

Types accepted: <pidType>internalId</pidType> or <pidType>externalId</pidType>

Values: internalId is a reference to the internalId of the contact. Reference to username and external username were not found to be applicable for update cases. The pid

internalId can be retrieved through the getSupplierContacts call with a reference to external username (as shown below).

Examples in get+updateSupplierContacts API call:

```
</request>
.....<pid>33621822</pid>
.....<pidType>internalId</pidType>
.....<sid>30001925</sid>
.....<sidType>internalId</sidType>
.....</updateRequest>
.....</sup:updateSupplierContacts>
<soapenv:Body>
.....<sup:getSupplierContacts2>
.....<request>
.....<pid>externalUsername@demo.com</pid>
.....<pidType>externalId</pidType>
.....<sid>30042285</sid>
.....<sidType>internalId</sidType>
.....</request>
.....</sup:getSupplierContacts2>
```

Conclusion

Please direct any requests, feedback, comments, and concerns to psintegrations@aravo.com.



The Definition of Better Business

Better business is built on acting with integrity.

It commands better performance, delivering better efficiency, collaboration, and financial outcomes.

It inspires trust.

But better business is more than that – it's about lifting the ethical standard of an entire business ecosystem to build a better world.

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